

Subrat Kumar Panda

Technology Analyst | Bhubaneswar

NodeJS, Multi Cloud, Microservices

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Professional Summary:

Senior Backend Engineer with nearly half a decade of experience in designing and scaling high-throughput, cloud-native backend systems across Digital Retail and Healthcare in Agile and Scrum environments. Specialized in Node.js, cloud platforms (AWS, Azure, IBM), microservices, & event-driven architectures, delivering measurable impact through cost optimization, performance improvements, and technical leadership. Proven experience building backend systems processing large data volume of transactions with zero data loss.

Skills:

- **Languages:** Node.js, Java, JavaScript, C++, SQL
- **Backend & Architecture:** Express.js, REST APIs, Microservices, Event-Driven Architecture, Distributed Systems, System Design
- **Databases:** PostgreSQL, MySQL, MSSQL, MongoDB, Redis
- **DevOps:** Docker, Kubernetes, CI/CD Pipeline design, Git, Azure DevOps
- **Observability:** New Relic, Mixpanel, CloudWatch, Grafana, Uptime Kuma
- **Cloud Platforms:**
 - **AWS:** ECS, Lambda, ELB, API GW, S3, SNS, SQS, CloudWatch
 - **Azure:** Azure AD, AKS, ACR, App Service, Functions Apps, Logic Apps, Key Vault, APP GW
 - **IBM:** Cloud Apps, IKS, IKP, Secrets Manager

Experience:

Technology Lead at Infosys Limited Bhubaneswar (Aug 2024 – present)

1. Application Development & RunOps Maintenance Project (Retail & Ecom - NodeJS, AWS, MongoDB, PostgreSQL):

- Lead backend development and RunOps support for a large retail MNC platform handling high-volume transactions.
- Developed the enhanced logic involved in self-checkout order services.
- Built a high-availability back-office store system using AWS ECS and RabbitMQ supporting 10M+ weekly transactions with zero data loss.
- Refactored delivery order flows and APIs, reducing response times and lowering production error rates significantly.
- Migrated APIs from AWS Lambda to ECS containers, improving response time by 50% and cutting costs by 30%.
- Reduced operational costs by up to \$5,000 per month by implementing New Relic pipeline filtering technique, eliminating unwanted ingested data.
- Revamped the LLD of integration service for third-party vendors to streamline inventory sync and order settlement thereby reducing errors by 20%.
- Optimized serverless APIs on AWS Lambdas and MongoDB by improving code quality, Redis caching, and event-driven design.
- Built proactive alerting and monitoring via CloudWatch, New Relic and Mixpanel, enabling early anomaly detection.
- Designed and implemented centralized observability using CloudWatch alarms, Log Insights dashboards, and custom metrics across compute services
- Developed automation scripts to design AWS CloudWatch metrics dashboards, accelerating dashboard setup from 2 weeks to under 10 mins
- Resolved critical bug fixes with major customer impact and Contributed to CAB releases, and API enhancements.
- Developed Node.js-based automation Jobs to generate and deliver daily data reports, enhancing visibility for stakeholders and core teams.

Systems Engineer at Tata Consultancy Services Hyderabad (Jul 2021 – Jul 2024)

1. Microservices Migration Project (Healthcare & Retail - NodeJS, Azure, MSSQL, PostgreSQL, MongoDB, Kubernetes):

- As the Offshore Technical Lead for a healthcare MNC, led the Intelligent Automations backend API team in developing & testing scalable systems.
- Led HLD & LLD redesign of a Node.js chatbot platform, migrating monolith APIs to microservices and achieving \$4,000/month cost savings.
- Developed and architected a multi-channel chatbot backend supporting 9 frontend platforms (Websites, MS Teams, SharePoint, Facebook Messenger, Workplace, Telegram, Twitter, Twilio SMS, WhatsApp) using the Adapter Design Pattern and NodeJS, with multiple third-party API integrations.
- Designed CI/CD pipelines for Docker containerization for an ACR, reducing manual efforts & AKS deployment time by approximately 15-20 hours monthly.
- Optimized SQL database usage through query optimization (Indexing & Restructuring Joins), decreasing API response time by 25%.
- Played a key role in enterprise-level migration from Kong to APIGEE API Gateway, improving security by 40% and performance by 33%.

2. Intelligent Automation Project (Bio Pharma & Life Science - NodeJS, Java, Azure, IBM Cloud, PostgreSQL):

- Developed a high-value chatbot and its Backend System, resulting in 2FTE savings & monthly cost reductions of \$15,000 for the client.
- Built advanced Java & Node - based RESTful APIs to serve as a backend service for RPA bots, handling over 2 million weekly transactions with 0 downtime.
- Optimized IBM & Azure cloud resources, leading to a cost savings of \$3,500 per month for the client.
- Developed automated audit reporting tools for RPA bot activities using Azure function apps.
- Automated Unit testing process with all possible edge cases using PowerShell scripts.
- Contributed to over 200+ code reviews, offering debugging guidance & addressing concerns regarding web applications and APIs utilized in RPA services.

Achievements & Accomplishments:

- Received the Rising Star, Insta & Multiple Client Awards at Infosys for driving innovation and impactful project contributions.
- Honored with the Service & Commitment Award, OnTheSpot Award, Applause Award, Star Team Award, Elevate Wings Award for dedicated service to TCS.
- Awarded as National Winner of 'Smart India Hackathon 2020' Hardware Edition (Dept. of Scientific & Industrial Research) by MHRD.
- Excelled as a National Finalist in 'L&T Techgrium 2019-20' & as an 'Emerging Innovator' by KPIT Technologies at 'KPIT Sparkle 2020'

Projects:

- **Pizza Store Project:** (Website : <https://pizza-website-delta-six.vercel.app/> & Repo : <https://github.com/subrat25/pizza-website>)
 - Developed a pizza store application with a Basic UI and a robust backend using Node.js, Express.js, and MongoDB
 - Implemented core features including user management, address handling, cart functionality, order processing, and email receipt generation
 - Designed and integrated request response loggers along with secure & encrypted authentication and authorization using JWT
- **Multiple Chatbot Backend:** (https://github.com/subrat25/test_chatbot_template.github.io)
 - Developed a backend using Node.js Express, integrating 2 Chatbot AI providers (Watson and Google Dialogflow)
 - Implemented robust database operations through Sequelize ORM, ensuring scalability and maintainability. & Enabled efficient logging of conversation details using MySQL and PostgreSQL, streamlining data management.

Certifications & Trainings:

- Upscaling expertise in Data Structures, Algorithms, Competitive coding and Software Development at Scaler Academy.
- Certified in AWS Cloud Technical Essentials Issued by Amazon Web Services via Coursera.
- Completed Certification courses about Building Chatbots using IBM Watson, Google Dialogflow & Azure (LinkedIn Learning & Cognitive Class)

Education:

Veer Surendra Sai University of Technology, Burla

B. Tech (Electrical & Electronics)

Year:2021

CGPA: 7.95